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Citation for published version:
Wang, D, Pang, W, Cao, Z, Song, L, An, L, Wu, X, Zhao, P, Wang, L & Zhou, Y 2025, 'A Lightweight Model LGCSPNet for Sitting Posture Risk Management Applications', *Expert Systems with Applications*, vol. 291, 128486. https://doi.org/10.1016/j.eswa.2025.128486

Digital Object Identifier (DOI):
10.1016/j.eswa.2025.128486

Link:
Link to publication record in Heriot-Watt Research Portal

Document Version:
Peer reviewed version

Published In:
Expert Systems with Applications

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Download date: 06. Aug. 2026

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A lightweight model LGCSPNet for sitting posture risk management applications

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Abstract

Current methods for sitting posture recognition typically follow a pipeline involving keypoint extraction and skeleton graph construction, followed by pose classification using Convolutional Neural Networks (CNNs) or Vision Transformers (ViTs). However, CNNs struggle to model long-range dependencies among keypoints, whereas ViTs suffer from high computational costs. Moreover, both approaches tend to introduce redundancy during feature modeling. To improve efficiency, some studies have explored direct classification using keypoint coordinates, but these methods often fail to balance high accuracy with computational efficiency. To this end, this paper proposes a new model LGCSPNet with lightweight graph convolution modules (LGC) and a contrastive learning module. Firstly, LGC enables efficient full-keypoint communication by shifting features across keypoint channels, allowing each keypoint to access global context at minimal computational cost. Building on this, LGC enhances sitting posture detection by computing 3D attention weights via a parameter-free energy function with a closed-form solution, enhancing feature learning for posturally significant keypoints. The contrastive learning module enhances differentiation between similar postures in different categories by strategically selecting feature samples. Experiments on public

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human posture datasets and our custom sitting posture dataset show that LGCSPNet has only 0.097M parameters while achieving a 99% recognition rate. It surpasses existing models in terms of parameter quantity and accuracy. Guided by ergonomic metrics, our model enables posture correction and mitigates long-term sitting-related injuries.

Keywords: Sitting posture recognition, Graph convolutional, Lightweight network, Ergonomics

1. Introduction

With the rapid development of the economy, prolonged sedentary work has become increasingly common in modern society. Statistics indicate that people now spend an average of seven hours per day sitting while working (Rosenkranz et al., 2020). For comfort, many individuals frequently adjust their sitting posture during long hours of work. However, these seemingly comfortable postures are often unhealthy. Prolonged maintenance of improper sitting postures can have severe health implications, frequently leading to issues such as muscle soreness, back pain, spinal pressure, and a range of other health issues (Feng et al., 2020). Office work is associated with a multitude of health issues, including musculoskeletal disorders (MSDs) and computer vision syndrome (Zhang et al., 2022), which are exacerbated by poor sitting habits and prolonged sedentary behavior.

To detect bad sitting postures in time and make improvement suggestions, researchers have proposed various posture monitoring methods. Among them, most approaches involve installing sensors to collect human posture data and detect postures (Rodriguez-Martin et al., 2013; Lin et al., 2022; Jian & Shijie, 2024; Gupta et al., 2020; Zhang et al., 2023; Yan et al., 2018b; Morales-Nolasco et al., 2024; Gupta et al., 2024; Yao et al., 2023; Tsai et al., 2023; Fan et al., 2022; Zaltieri et al., 2022). However, these methods tend to be highly intrusive, not only causing discomfort to the monitored individuals but also requiring expensive sensor equipment that is often inconvenient to install. In contrast, computer vision-based posture analysis methods require only cameras to capture user postures, which are analyzed using deep learning models (Ding et al., 2024; Lafuente-Arroyo et al., 2022; Zhou et al., 2024; Ogundokun et al., 2022; Pradeep & Sinha, 2024; Carneros-Prado et al., 2024; Hoefflin et al., 2024; Zhao & Su, 2024; Lin et al., 2023; Kulikajevas et al., 2021; Cao et al., 2025). Such approaches are less intrusive, easier to operate,

and increasingly suitable for applications in daily life. To this end, commonly used deep learning models include convolutional neural networks (CNNs) (Li et al., 2021) and Vision Transformers (Han et al., 2022). However, both models face significant challenges in sitting posture recognition tasks. CNNs and Vision Transformers heavily rely on the global structure of images, whereas the skeletal morphology of the human body is the critical feature for sitting posture recognition. Processing the complete spatial structure of an image each time not only increases the computational burden of the model but also impacts its runtime speed, hindering lightweight model design and leading to inefficient use of computational resources. The convolution operation in CNNs is limited to local receptive fields, restricting their ability to capture long-range dependencies. This is particularly problematic when processing skeletal data, as connections between distant body parts are often overlooked. Although Vision Transformers utilize self-attention mechanisms to capture global dependencies, their high computational complexity makes them less suitable for lightweight models. Moreover, existing methods primarily focus on extracting local features within individual samples, while neglecting the rich interrelationships between different sample categories.

In summary, existing deep learning-based sitting posture recognition methods face three main issues: First, the models have large parameter sizes and slow inference speeds, making them unsuitable for deployment on resource-constrained edge devices. Second, these models have limited receptive fields, making them difficult to flexibly capture relationships between distant body parts. Third, these models focus excessively on the local features of individual postures, neglecting the connections between different samples.

To address these key issues, we propose a novel model named LGCSP-Net, which integrates lightweight graph convolution (LGC) modules and a contrastive learning module. The LGC module facilitates efficient communication across all keypoints by shifting features among keypoint channels, thereby enabling each keypoint to capture global context with minimal computational overhead. Furthermore, LGC employs a parameter-free energy-based mechanism with a closed-form solution to compute 3D attention weights, leading to more accurate sitting posture detection and improved feature learning for posture-critical keypoints. In parallel, the contrastive learning module enhances the model's ability to distinguish between visually similar postures from different categories by strategically sampling feature representations.

It is worth emphasizing that the proposed algorithm does not rely on

multi-view inputs to function effectively. Multi-view datasets are only used to evaluate the model's robustness under complex viewpoint changes, our model can achieve reliable recognition results using single-view images captured by common devices such as laptop webcams or smartphone cameras. This significantly lowers hardware and deployment costs, and enables a non-invasive, lightweight, and portable solution suitable for daily-life scenarios without requiring additional sensors or environmental modifications. Compared to traditional systems that depend on pressure sensors or wearable devices, our approach offers better usability and broader adaptability in real-world applications.

In summary, our main contributions are as follows:

- To address the high computational cost and limited effectiveness of current sitting posture recognition methods in modeling long-range keypoint dependencies, we propose a lightweight graph convolution module (LGC), which leverages cross-channel feature transfer and a closed-form attention mechanism based on a parameter-free energy function to enable efficient dependency modeling and enhanced perception of critical regions with minimal parameters.

- To enhance the model's ability to distinguish between similar postures and better discriminate boundary samples, we introduce a contrastive learning mechanism. By strategically selecting the most confusing samples during training, the model significantly improves its recognition of subtle posture differences.

- We design a novel Lightweight Graph Convolutional Comparison Sitting Posture Recognition Network LGCSPNet, and we conduct extensive experiments on both public datasets and our own dataset. Experiments on public human posture datasets and our custom sitting posture dataset show that LGCSPNet has only 0.097M parameters while achieving a 99% recognition rate. It surpasses existing models in terms of parameter quantity and accuracy.

- To address the limited generalization ability of existing sitting posture recognition models under varying viewpoints, we design and construct a multi-view sitting posture dataset. The dataset contains 49,798 images covering 7 posture categories, 7 static viewpoints, and 1 dynamic viewpoint. To the best of our knowledge, this is the first dataset in the

sitting posture recognition domain that includes such comprehensive viewpoint diversity, making it suitable for evaluating model robustness and generalization in complex multi-view scenarios. Dataset link: https://github.com/LKW123800/multi-view-sitting-posture-dataset

- To integrate sitting posture recognition with practical health applications, we draw inspiration from commonly used ergonomic risk classification principles to categorize the seven identified postures into high, medium, and low risk levels. Based on the recognition results, we provide actionable and user-friendly health recommendations to help mitigate the risks associated with prolonged sitting behavior.

The remainder of this paper is organized as follows. Section 2 reviews related work in the field. Section 3 provides a detailed description of the proposed LGCSPNet architecture, as well as the predefined sitting posture categories and their corresponding risk levels. Section 4 presents the constructed datasets, experimental setup, and comprehensive evaluation results. Section 5 discusses the strengths and limitations of the model in terms of feature representation, global perception, and real-world applicability. Finally, Section 6 concludes the paper by summarizing the key contributions and outlining future directions, including the development of more diverse datasets and the integration of personalized intervention mechanisms.

2. Related Work

Chen (2019) proposed a classroom student posture recognition system based on OpenPose (Cao et al., 2017), which used monitoring devices to detect student postures and extract pose features. The system employed a convolutional neural network (CNN) built with Keras for training, achieving a recognition accuracy of over 90%. Furthermore, Liu et al. (2020a) introduced a skeleton-based posture recognition framework, 3D PostureNet, which enhanced the robustness of posture recognition through skeleton data normalization and Gaussian voxelization. The framework employed a three-dimensional convolutional neural network (3D CNN) for end-to-end recognition and has demonstrated excellent performance on multiple public datasets. Although Liu's method performed well in multi-dataset tests, its computational complexity with 3D convolutions posed challenges for real-time applications due to higher computational demands. In addition, CNNs struggled

to capture long-range dependencies. To further optimize recognition accuracy and efficiency, Fang et al. (2022) proposed a novel Vision Transformer structure named DuFormer. This model enhanced positional information through dual-position embedding and integrated convolution with external attention mechanisms to construct the SPRNet model. During testing, SPRNet successfully identified three common postures and outperformed other state-of-the-art methods. Despite its high accuracy, the complexity of the Transformer structure leads to increased dependency on hardware resources, potentially raising deployment costs. In addition, Rodrigues et al. (2022) proposed a non-contact posture monitoring method based on Kinect cameras and computer vision algorithms, which evaluated the posture of office workers by estimating joint angles. This method showed a high correlation with the results from the goniometer, but it was computationally complex, had a longer processing time, and its real-time monitoring efficiency might be limited. Li et al. (2023) proposed a model for abnormal sitting posture recognition based on multi-scale spatiotemporal features of skeleton graphs. However, due to the heavy emphasis on temporal dynamics, the model introduced redundancy, which undermined its efficiency in recognizing static sitting postures. Cao et al. (2025) proposed a sitting posture recognition model with multi-view recognition capability. However, the algorithm for processing multi-view data increased the complexity of the system, and the acquisition of multi-view data also raised the deployment cost of the system. Although significant progress has been made in posture recognition using computer vision, existing methods often struggle to strike a good balance between capturing distal joint dependencies and model lightweighting. To address this, the proposed LGC module enables long-range joint interaction by shifting keypoint features across channels, effectively propagating global contextual information without incurring the cost of dense spatial operations. In parallel, a closed-form attention mechanism selectively enhances posture-critical joints without introducing learnable parameters, together achieving efficient yet expressive posture representation.

In recent years, research on lightweight network architectures has provided valuable insights for the efficient deployment of posture recognition models. Iandola (2016) proposed SqueezeNet, which introduced the "fire module" to reduce the number of convolutional kernels and replace traditional 3×3 convolutions with 1×1 convolutions, significantly compressing the model size. Chollet (2017) proposed Xception, a network architecture based on depthwise separable convolutions, which further enhanced the

model's nonlinear representation capability. Howard et al. (2019) introduced MobileNetV3, which combined depthwise separable convolutions with search optimization techniques, improving network performance while optimizing computational efficiency and accuracy. Han et al. (2020) proposed GhostNet, which utilized Ghost modules to generate features at low computational cost, greatly improved the computational efficiency of the network. Vasu et al. (2023) proposed MobileOne, an efficient neural network backbone optimized for mobile devices, which achieved extremely low latency and high accuracy on mobile devices by analyzing and optimizing the bottlenecks of existing networks. Although significant progress has been made in lightweight network design, most existing lightweight architectures improve computational efficiency by modifying standard convolutional structures. Representative approaches, such as depthwise separable convolutions and group convolutions, decouple the convolution operation into more efficient sub-processes. By reducing full connectivity across channels, these methods significantly lower the number of parameters and floating-point operations. However, this structural simplification inevitably weakens the model's ability to jointly capture spatial and channel-wise information, limiting its capacity to model complex structural dependencies. To address this limitation, we propose the LGC module, which leverages channel features to compute inter-keypoint relationships for guiding spatial modeling, while weighting channel features with inter-keypoint relationships before channel fusion, enabling mutual interaction to achieve joint spatial-channel modeling, effectively enhancing representational capacity and overcoming the shortcomings of traditional lightweight networks in joint modeling.

3. Method

In this section, we first provide an overview of the proposed human skeleton modeling and LGCSPNet, as shown in Fig. 1. Subsequently, we present a detailed explanation of each component of the LGCSPNet architecture in separate subsections.

3.1. Human Skeleton Modeling and LGCSPNet Network

We directly model the relationships within the human skeleton as a graph structure, where the joints are represented as the nodes of the graph, and the limb connections are represented as the edges. Specifically, we model a human skeleton as a set $G = (V, M)$, where $V = \{v_1, v_2, v_3, \ldots, v_n\}$ represents

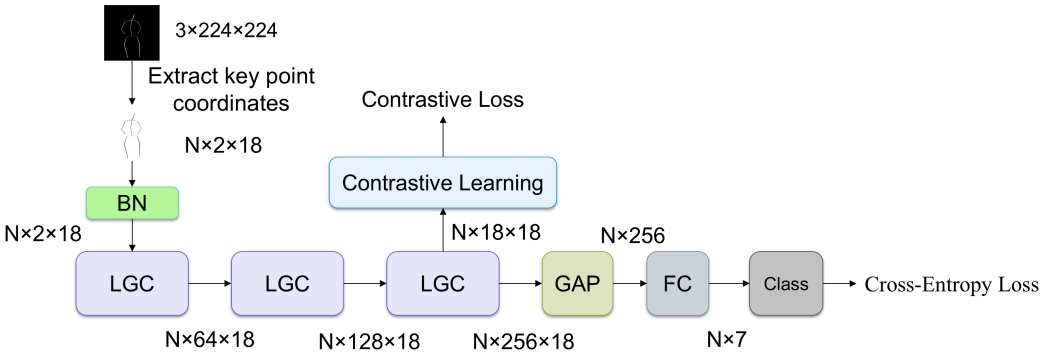

Fig. 1. Simple Structure of the LGCSPNet, BN represents batch normalization, GAP represents global average pooling, FC represents fully connected layer, and Class represents the classification result.

the n nodes of the skeleton, and $M \in \{0,1\}^{N\times N}$ is the adjacency matrix that encodes the connections between the nodes.

The architecture of the proposed LGCSPNet network is illustrated in Fig. 2. The LGCSPNet framework consists of a feature processing stage with multiple LGC modules and a contrastive learning component. First, the human skeletal keypoint data are input into the feature processing stage. After passing through three LGC modules, the number of channels in the model is adjusted from 2 to 64, 128, and 256, respectively, extract rich feature representations, yielding robust abstractions. The final LGC module integrates with the contrastive learning component, which strategically selects feature samples to construct representative positive and negative pairs. By optimizing with contrastive loss, the model enhances its discriminative power and generalization. LGCSPNet achieves a balance between model lightweightness and high recognition accuracy, effectively overcoming receptive field limitations and reducing computational complexity, making it well-suited for sitting posture recognition tasks.

3.2. Lightweight graph convolution core

Existing mainstream approaches still struggle to model long-range dependencies between keypoints effectively. Due to their local receptive fields, Convolutional Neural Networks (CNNs) inherently struggle to capture relationships among distant keypoints. Although Vision Transformers (ViTs) can model global dependencies, their high computational cost—particularly with high-resolution inputs—limits their use in real-time or resource-constrained

scenarios. Additionally, both methods often introduce redundant information during feature extraction, leading to inefficiency. To enhance computational efficiency, some recent studies have explored pose modeling directly from keypoint coordinates. While these methods show promise, they still fail to balance high accuracy with low computational cost.

Building on cross-keypoint feature transfer (Cheng et al., 2020) and energy-function-based attention (Yang et al., 2021), we propose the Lightweight Graph Convolution Core (LGC), which efficiently models global keypoint dependencies while enhancing pose-critical feature representations. Specifically, the proposed LGC module uses channel features to model inter-keypoint relationships, guiding spatial modeling. It also weights channel features with these relationships before channel fusion, enabling joint spatial-channel modeling and enhancing representational capacity.

The LGC module starts with a 1×1 convolution to adjust the channel dimension of the input feature map to boost representational capacity. The transformed features are then fed into two parallel branches of non-local shift graphs (NSG). Using predefined index arrays, NSG rearranges features along the channel dimension, allowing each keypoint to capture information from all others. The outputs of the two branches form an adaptive adjacency matrix, which is normalized and processed by a Softmax function to produce a spatial attention map capturing node interactions. The attention map is weighted and combined with the input features, with a 1×1 convolution integrating spatial and channel information. The enhanced features are refined by SimAM, using a closed-form energy function to emphasize critical keypoints for pose estimation while suppressing irrelevant ones. Finally, a residual connection combines the original and enhanced features, producing a contextually rich and coherent representation.

To elaborate, the NSG operation, as illustrated in Fig. 3, represents a single-frame skeleton as $F \in \mathbb{R}^{N \times C}$, where N denotes the number of skeletal nodes and C represents the channel dimension. For the i-th node, in the j-th channel, the node is shifted by a distance of $j \ mod \ N$, and the other nodes are shifted sequentially to the channel position of this node. This process is iterated until all channels are processed. The NSG process with forward cyclic shifting can be expressed using the Equation (1), where $\tilde{F}[i, j]$ represents the information at the j-th channel of the i-th node after NSG.

$$\tilde{F}[i, j] = F(((i - (j \bmod N)) \bmod N), j) \tag{1}$$

In the LGC module, the input data X first pass through a 1×1 con-

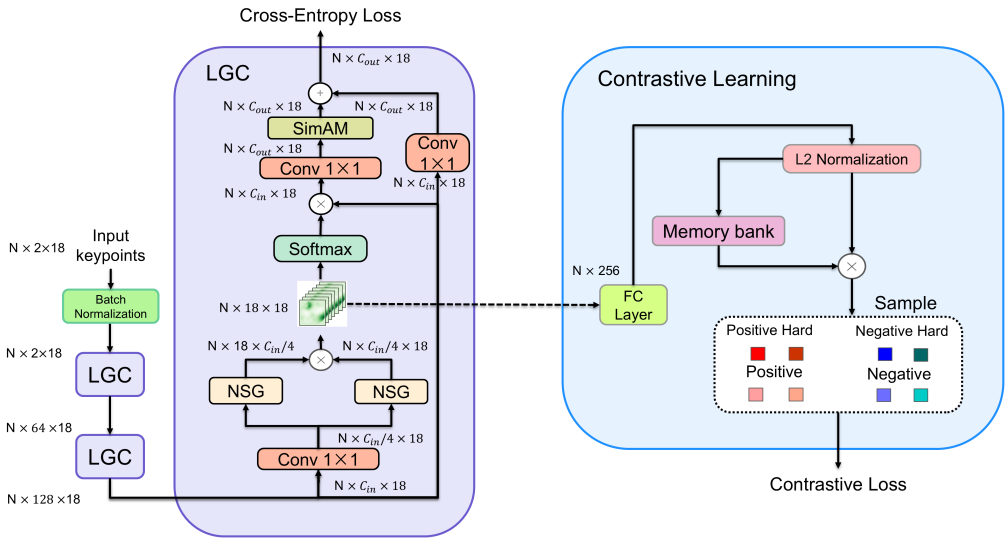

Fig. 2. The overall structure of the LGCSPNet. $\oplus$ denotes the element-wise addition, $\otimes$ denotes the matrix product, and FC layer refers to the fully connected layer. The three-layer LGC modules progressively expand the number of channels, increasing from 2 to 64, 128, and finally 256.

volution layer to adjust the number of channels as shown in Equation (2). The processed data X_1 are then fed into two non-local shift graph operations, which perform shifts in different directions: one applies a forward cyclic shift, while the other applies a backward cyclic shift. This approach generates two distinct feature maps X_l and X_r, as shown in Equations (3) and (4) .

$$X_1 = \text{Conv1}\times1(X) \tag{2}$$

$$X_l = \text{NSG}^{+}(X_1) \tag{3}$$

$$X_r = \text{NSG}^{-}(X_1) \tag{4}$$

Finally, these two feature maps X_l and X_r are matrix multiplied, resulting in a unique graph G corresponding to each data point as shown in Equation (5).

$$G = X_l \times X_r \tag{5}$$

The generated unique graph G undergoes a softmax operation as shown in Equation (6) and is then matrix multiplied with the skip-connected initial input X to obtain the enhanced feature representation Z as shown in Equation

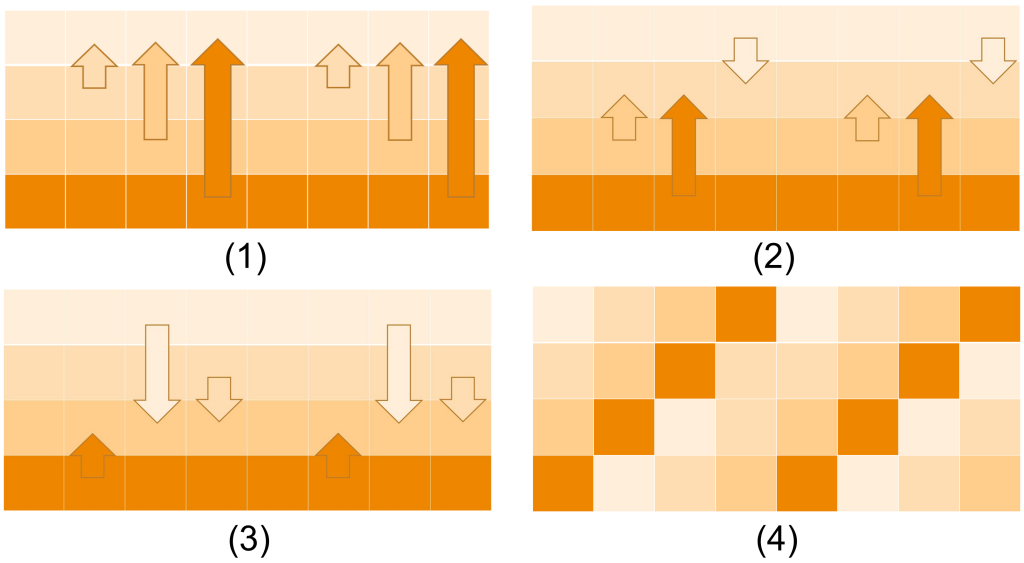

Fig. 3. Illustration of the non-local graph shift operation. Each row corresponds to a different keypoint, and each column corresponds to a different channel. (1) shows how the features of keypoint 1 are shifted across channels. (2) shows the same operation applied to keypoint 2. (3) applies the shift to keypoint 3. (4) presents the final result after all keypoints have undergone the shift. This shift operation enables each keypoint to receive and integrate information from other keypoints, thereby facilitating global information exchange across the graph.

(7).

$$Graph = \text{Softmax}(Graph) \tag{6}$$

$$Z = Graph \times X \tag{7}$$

It is noted that the NSG introduces no additional learnable parameters, maintaining the lightweight design of the model. By leveraging the NSG, information exchange is facilitated between previously unconnected nodes based on the physical connections of the human body. This allows the model to comprehensively utilize all joint information. The NSG not only reduces computational complexity but also effectively addresses the traditional network models' limitations in capturing limb dependencies within skeletal graphs. As a result, it significantly enhances the model's performance in sitting posture recognition tasks.

In the NSG, the limb connection weights between different nodes are identical. However, the importance of different body parts varies across different postures. To better capture this variation, we integrate a parameter-free attention module, SimAM (Yang et al., 2021), into the LGC module. This module computes an attention map that reflects the varying weights of different body parts, allowing the model to adaptively focus on the most critical regions for posture recognition.

Existing attention modules typically compute attention independently along either the spatial or channel dimension, and they often rely on heuristic designs. This leads to limited global context, suboptimal information utilization, and the introduction of additional parameters to the model. SimAM (Yang et al., 2021) addresses this issue by drawing inspiration from the principles of brain neurons, using linear separability to compute a unified 3D attention map via an energy function without introducing learnable parameters. This makes SimAM a parameter-free, lightweight attention module. The energy function for each neuron can be expressed as shown in Equation (8).

$$e_t(w_t, b_t, y, x_i) = \frac{1}{M-1} \sum_{i=1}^{M-1} \left(-1 - (w_t x_i + b_t)\right)^2 + \left(1 - (w_t t + b_t)\right)^2 + \lambda w_t^2 \tag{8}$$

In the above, t and x_i represent the target neuron and other neurons within each channel, respectively. w_t and b_t denote the transformation weights and biases. $M = H \times W$ represents the number of neurons within a single

channel, while λ is the regularization parameter. The expressions for w_t and b_t are as shown in Equations (9) and (10).

$$w_t = -\frac{2(t-\mu_t)}{(t-\mu_t)^2 + 2\sigma_t^2 + 2\lambda} \tag{9}$$

$$b_t = -\frac{1}{2}(t+\mu_t)w_t \tag{10}$$

Among them, μ_t and σ_t^2 represent the mean and variance of neurons excluding t. To simplify the calculation, it is assumed that neurons within the same channel are identically distributed. Thus, the minimum energy can be calculated as shown in Equation (11).

$$e_t^* = \frac{4(\hat{\sigma}^2 + \lambda)}{(t-\hat{\mu})^2 + 2\hat{\sigma}^2 + 2\lambda} \tag{11}$$

The equation measures the linear separability between the target neuron and other neurons. The smaller e_t^*, the greater the linear separability between the target neuron and other neurons, and the higher the weight of the target neuron should be. Therefore, the weight of the target neuron is given by $\frac{1}{e_t^*}$. In this way, the SimAM module achieves the calculation of attention weights without introducing additional learnable parameters, ensuring lightweight computation.

The enhanced feature representation Z enters SimAM after 1×1 convolution adjustment. Subsequently, SimAM is used to further refine Z, as shown in Equation (12).

$$Z = \text{SimAM}(\text{Conv}1\times1(Z)) \tag{12}$$

After that, BatchNorm is applied to Z as shown in Equation (13) to stabilize the training process and improve generalization. Finally, element-wise addition is used to fuse the enhanced and refined features Z with the input X adjusted by 1×1 convolution to generate the intermediate representation Y, which will pass through the ReLU activation function to introduce nonlinearity and generate the output of the module as shown in Equations (14) and (15).

$$Z = \text{BatchNorm}(Z) \tag{13}$$

$$Y = \text{Conv}1\times1(X) + Z \tag{14}$$

$$Y = \text{ReLU}(Y) \tag{15}$$

3.3. Contrastive Learning Module

The LGC module demonstrates its ability to effectively extract and abstract features within individual samples. However, we believe that abundant inter-sample category relationships remain unexplored. Therefore, we introduce contrastive learning (Huang et al., 2023) and apply it to sitting posture recognition. To embed graphs into a vector space and perform contrastive learning in a unified feature space, the input data to the contrastive learning module are processed using spatial displacement graph operations within the LGC module. Specifically, two feature maps are generated by performing forward cyclic shift and backward cyclic shift, respectively. These two feature maps are then multiplied to produce a unique vector corresponding to each input datum.

The contrastive learning module includes a memory bank that stores the unique vectors generated for each input data point after the graph-shifting operations performed by the LGC module. The contrastive learning module constructs positive and negative sample pairs based on the labels of each graph. For contrastive learning, there exists a positive sample pair I and I^+ with the same label, along with their corresponding embedding vectors f and f^+. A negative sample I^- is selected from the negative set N^-. This sample I^- has a label that differs from the current sample and is associated with the embedding vector f^-. The similarity can be computed as shown in Equation (16):

$$\text{sim}(f^+, f^-) = \frac{f^+ \cdot f^-}{\|f^+\|\|f^-\|} \tag{16}$$

In contrastive learning, the commonly used InfoNCE loss function can be expressed as shown in Equation (17):

$$\mathcal{L}_{NCE} = -\log \frac{\text{sim}(f, f^+)/\tau}{\text{sim}(f, f^+)/\tau + \sum_{f^- \in N^-} \text{sim}(f, f^-)/\tau} \tag{17}$$

here $\tau > 0$ is the temperature parameter.

Positive and negative examples are sampled from the memory bank to construct sample pairs. Specifically, when selecting a vector v as the anchor sample, to improve the quality of sample pairs and avoid selecting overly simple samples, the selection is based on similarity $sim(f^+, f^-)$. Select a portion of negative samples v^- with higher similarity, which are the most difficult negative samples to distinguish. Similarly, select a portion of positive samples v^+ with lower similarity, which are the most difficult positive

samples to distinguish. At the same time, to increase sample diversity, a certain number of negative samples are randomly selected as part of the negative sample pairs. This sample selection strategy effectively enhances the effectiveness and diversity of training, enabling the model to better learn the ability to distinguish between different samples.

For the entire memory bank, the loss function can be expressed as shown in Equation (18):

$$\mathcal{L}_{\mathrm{NCE}}^{\mathrm{Ins}} = -\sum_{v^+ \in N^+} \log \frac{\mathrm{sim}(v, v^+)/\tau}{\mathrm{sim}(v, v^+)/\tau + \sum_{v^- \in N^-} \mathrm{sim}(v, v^-)/\tau} \tag{18}$$

The total loss of the network can be expressed as shown in Equation (19):

$$\mathcal{L} = \mathcal{L}_{\mathrm{NCE}}^{\mathrm{Ins}} + \mathcal{L}_{CE} \tag{19}$$

In the above, L_{CE} represents the cross-entropy loss function of the LGC module. It is worth noting that the contrastive learning module is only activated during the training phase and is removed during the testing phase.

3.4. Predefined sitting posture categories

Based on existing ergonomic research, while considering common sitting postures and the associated health risks, we select seven sitting postures for study as shown in Fig. 4. These seven sitting postures also constitute the primary poses in the subsequent major research dataset, MSP. Back: The body leans back against the chair, with a lack of lumbar support, and legs naturally extended. Cheek: The hand supports the head, with the body slightly leaning forward. Down: The body leans forward significantly, with the head facing downward. Left: The body tilts to the left side. Leg: The legs are crossed. Right: The body tilts to the right side. Upright: The body sits upright, with the back close to the chair back and both feet flat on the floor.

To better manage health, and inspired by commonly used ergonomic risk classification principles, we categorize these postures into three levels: high, medium, and low, as shown in Table 1. Specifically, the classification is based on the deviation of each posture from the ideal neutral position and the potential damage it may cause to bones and muscles. The low-risk level is defined as a relatively healthy and friendly sitting posture. If maintained for a long time, it will not bring an obvious burden or discomfort to the body.

Table 1
Posture risk classification.

Posture	Risk Level
back	High
cheek	Medium
down	High
left	Medium
leg	Medium
right	Medium
upright	Low

Ergonomics believes that this kind of sitting posture is more conducive to the body to maintain a natural balance. The medium-risk level is defined as a sitting posture with certain potential risks in ergonomics. Although it may not be a big problem to adopt it for a short time, if it is maintained for a long time or habitually adopted, it may bring some potential negative effects to the body. Slowly accumulating may cause discomfort and even increase some health risks. High-risk level is defined as a type of sitting posture that is considered very unhealthy in ergonomics and needs to be corrected immediately. The characteristics of this type of sitting posture are: they are more obviously deviated from the natural physiological structure of the human body, the body is in a state of tension and pressure, and the possibility of musculoskeletal problems is greatly increased.

Upright is classified as a low risk level. In this level of sitting posture, the human body remains naturally sitting on the chair, neither leaning forward nor backward, with both feet on the ground, and the spine maintains a natural curve. Low-risk sitting posture can disperse pressure, reduce unnecessary muscle tension, and is generally more friendly to the body (Harrison et al., 1999). Leg, left, right, cheek, and other postures are classified as medium risk. The risk of this level of posture is mainly that these postures will put more pressure on the spine to a certain extent, or affect local blood circulation, increase hip pressure, and cause pelvic tilt (Yu & An, 2015; Marras & Granata, 1997). Although it is not an immediate and serious damage, it may appear over time. Therefore, for medium-risk sitting postures, our advice is to minimize long-term maintenance and consciously adjust the posture to a low-risk posture, get up and move around appropriately, and give the body a chance to relax and recover. Back and down is classified as high risk. These

postures will put the spine, muscles, and other tissues in an extremely unnatural and uneven state of force. In the long run, it is easy to cause health problems such as low back pain, cervical problems, etc (Waongenngarm et al., 2015). For high-risk sitting postures, our attitude is to avoid them as much as possible. If such a posture occurs, it should be corrected immediately and adjusted to a proper sitting position.

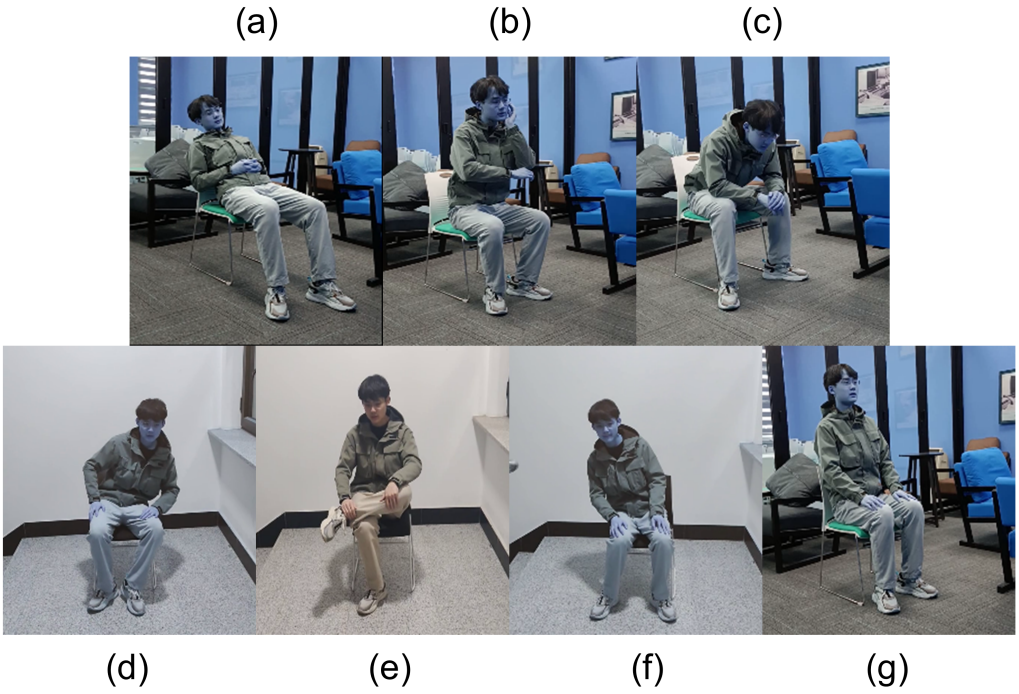

Fig. 4. Examples of the 7 selected sitting postures. (a): leaning backward (back), (b): chin on hand (cheek), (c): leaning down (down), (d): leaning left (left), (e): crossing leg (leg), (f): leaning right (right), (g): upright sitting (upright).

3.5. Recommendations for Sitting Posture Correction

To extend the proposed sitting posture recognition model into a more proactive health intervention tool, we developed a dedicated recommendation module that works in conjunction with the proposed sitting posture recognition model to provide corrective guidance. The process is shown in the Fig. 5. This component processes the classification results from the recognition model, identifying the type of incorrect posture and generating corresponding ergonomic suggestions.

At the core of the recommendation module lies a purpose-built knowledge base comprising three key components: (1) The set of standard healthy posture categories identified by the proposed model, each associated with a defined risk level; (2) The set of abnormal posture categories identified by the proposed model, each similarly annotated with its corresponding musculoskeletal risk level and (3) a comprehensive set of clear, actionable corrective instructions tailored to each identified abnormal posture. Notably, the posture categories and associated risk levels within the knowledge base are fully aligned with the classification schema and risk stratification framework defined in Section 3.4.

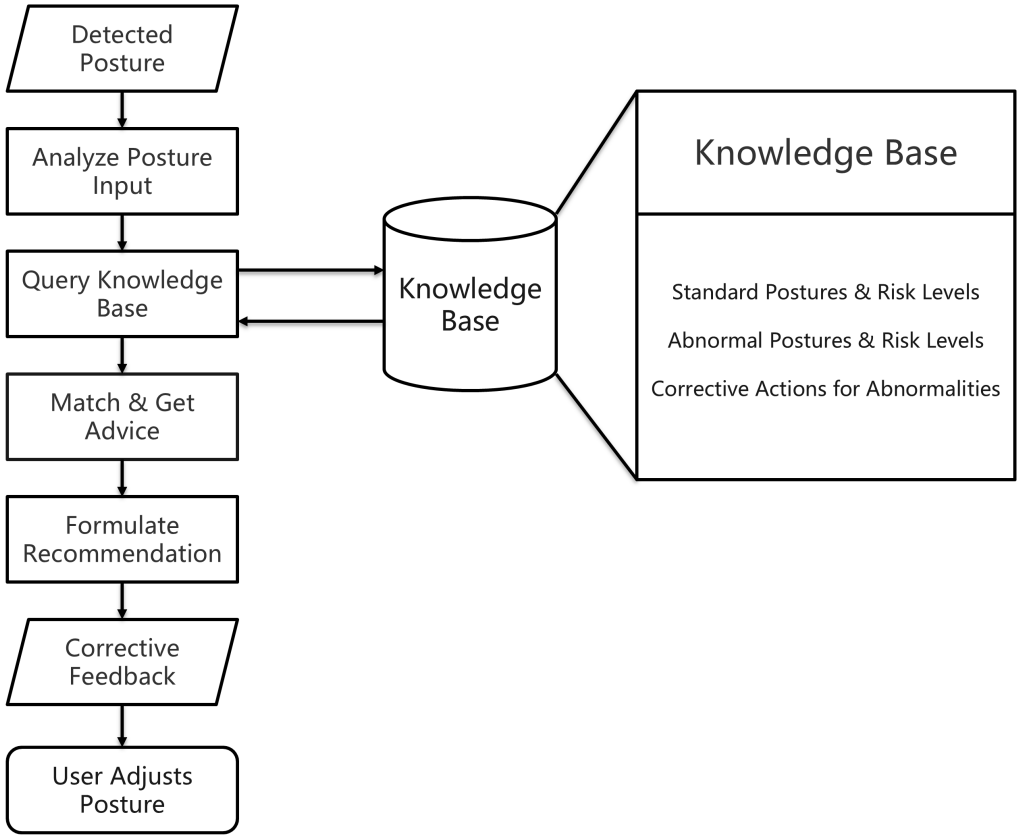

Fig. 5. Flowchart of recommendations for sitting posture correction.

The recommendation logic is structured around a rule-based framework, operating on an "IF [specific incorrect posture detected], THEN [issue corresponding corrective instruction]" principle. This ensures feedback is both

timely and context-sensitive. Recommendations are delivered through non-intrusive channels, primarily visual cues such as brief on-screen text or illustrative icons, with optional auditory alerts to enhance user awareness.

By integrating this recommendation component, the model transitions from passive monitoring to active user support, facilitating the development of healthier sitting habits. In addition to addressing immediate postural deviations, it fosters long-term ergonomic awareness and sustainable behavioral change.

4. Experiments

In this section, we provide a detailed description of the datasets, parameter settings, experimental details, and final results. The experiments are conducted on an NVIDIA GeForce RTX 2080 Ti GPU, and the proposed network is trained on a public body pose dataset as well as a self-constructed proprietary dataset, demonstrating excellent performance. To ensure a fair comparison with other models, we train and test other mainstream network models on the same datasets and hardware environment. Experimental results demonstrate that the proposed model outperforms the compared models in terms of lightweight design and inference accuracy, demonstrating its advantages in both performance and efficiency.

4.1. Datasets

Multiview sitting posture dataset. This study constructs a multi-viewpoint sitting posture recognition image dataset, aiming to address the limitations of existing sitting posture datasets in terms of viewpoint diversity. The data are collected in an indoor office environment using a monocular camera, and the most representative typical postures of male subjects in seven sitting states are collected from eight viewpoints, including seven fixed viewpoints and one dynamic viewpoint: back, cheek, downward, left, leg, right, and upright. The posture descriptions are shown in Table 2. These categories are determined by searching ergonomic literature related to sitting posture and sitting musculoskeletal problems (Vergara & Page, 2002; Gallagher, 2005; Kee & Karwowski, 2001; Scena & Steindler, 2008). After rigorous manual screening, the dataset is established, which eventually contains 49,798 images. We believe that this dataset can provide a more comprehensive and reliable multi-viewpoint data resource for sitting posture recognition research. Given the large scale of the dataset, we select data

from three key viewpoints: the front view (A), the left-front-side view (B), and the right-front-side view (C), as shown in Fig. 6.

Table 2
Posture description.

Posture	Description
back	lean back
cheek	chin on hand
down	lean forward and bend over
left	body leans to the left
leg	cross one leg over the other
right	body leans to the right
upright	relax and sit up straight

In the data construction process, we design two different dataset configurations by combining viewpoints in distinct ways. First, images from Viewpoint A and Viewpoint B are mixed and then divided into training, validation, and test sets, forming Dataset I (MSP I), as shown in Table 3. Second, data from Viewpoint A and Viewpoint B are mixed and split only into training and validation sets, while data from Viewpoint C are kept separately as the test set, forming Dataset II (MSP II), as shown in Table 4. Dataset I is primarily used to verify the model's recognition performance under identical viewpoint conditions and to validate the effectiveness of the proposed module. Dataset II is designed to evaluate the model's generalization ability in cross-viewpoint scenarios.

Table 3
A summary of multiview sitting posture dataset I.

Perspective	Train	Test	Val
A	3,563	762	770
B	3,398	726	734
A+B (Total)	6,961	1,488	1,504

Body pose dataset. To further validate the effectiveness of our method, we also conduct experiments on the public Body pose dataset (Hidalgo & Casas, 2015) (BPD). This dataset contains 10 static body poses of varying complexity performed by 12 volunteers, totaling 7,981 samples. To ensure compatibility with our model setup, we use OpenPose to detect key points

Table 4
A summary of multiview sitting posture dataset II.

Perspective	Training	Test	Val
A	3,563	0	770
B	3,398	0	734
C	0	1,440	0
A+B+C (Total)	6,961	1,440	1,504

in the sample images of the dataset, obtaining 2D coordinates for 18 key points per sample. The dataset is partitioned as follows: data from the first 8 volunteers are used as the training set, data from the last 2 volunteers as the validation set, and the remaining data as the test set to evaluate model performance. In the experiments, the 2D coordinates of the key points are provided as input to the model. By analyzing these input features, the model is able to capture the geometric properties of the poses, thereby achieving robust performance evaluation.

Data processing. This study uses OpenPose (Cao et al., 2017) to extract 18 skeletal keypoints from the original images. For the non-keypoint-based model, we construct skeleton maps by removing redundant background based on the extracted keypoints, and normalize them into sample images of size 224×224 pixels as model input. For the keypoint-based model, we normalize the coordinates of the 18 keypoints extracted by OpenPose to the range [0, 1] and use them as model input.

4.2. Experimental Settings

In this study, all experiments are conducted on a Linux platform using the PyTorch 1.12 deep learning framework with Python 3.8. The hardware setup consists of a NVIDIA GeForce RTX 2080 Ti GPU and a 12th Gen Intel (R) Core (TM) i7-12700F CPU. The training configuration was carefully optimized for each dataset to evaluate model performance under the same hardware resources. Table 5 lists the detailed training parameters applied to each dataset.

4.3. Comparison with Other Network Models

To comprehensively evaluate the performance of our proposed sitting posture recognition model, we select a diverse set of representative deep learning models for comparison. These include classical CNN architectures such as

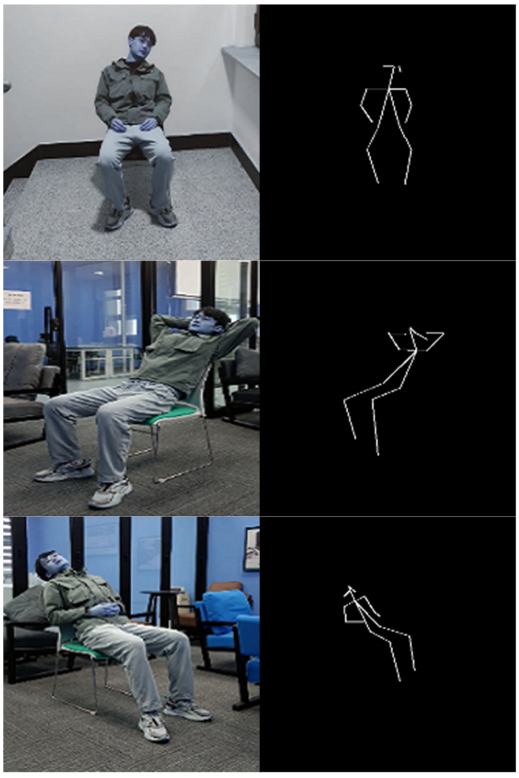

Fig. 6. Partial data from perspectives A, B, and C.

Table 5
Training hyperparameters used in our experiments.

Hyperparameter	MSP I	MSP II	BPD
Optimizer	SGD	SGD	SGD
Base learning rate	0.44	0.44	0.629
Weight decay	0.0004	0.0004	0.0004
Optimizer momentum	0.9	0.9	0.9
Learning rate schedule	Step Decay	Step Decay	Step Decay
Decay step (epochs)	35	35	35
Learning rate decay factor	0.1	0.1	0.1
Warmup schedule	Linear	Linear	Linear
Warmup epochs	5	5	5
Batch size	256	512	512
Num epoch	50	50	50

VGG16 (Simonyan, 2014), ResNet (He et al., 2016) and Xception (Chollet, 2017), lightweight networks such as MobileNetV3 (Howard et al., 2019), SqueezeNet (Iandola, 2016), LMSPNet (Jiao et al., 2023) and GhostNet (Han et al., 2020), graph-based models like ST-GCN (Yan et al., 2018a) and MS-G3D (Liu et al., 2020b), and Transformer-based architectures including ViT-B/16 (Dosovitskiy, 2020) and Swin-T (Liu et al., 2021). Furthermore, we include hybrid models that combine the strengths of CNNs and Transformers, such as the CMT series (Guo et al., 2022), and newly proposed efficient architectures like the VanillaNet series (Chen et al., 2024), to ensure a comprehensive and fair evaluation.

All models are trained and tested on both the MSP I and the public BPD, as shown in Table 6 and Table 8. The results demonstrate that our model outperforms other models in terms of recognition accuracy while maintaining a smaller size.

To validate the cross-viewpoint generalization capability of the proposed model under extreme conditions, we conduct multi-viewpoint sitting posture recognition experiments on the multiview sitting posture dataset II. In this setting, the model is trained using data from viewpoints A and B, while being tested exclusively on viewpoint C, which is completely unseen during training. This deliberate design introduces a significant domain gap between training and testing, making the task considerably more challenging. As such, the lower performance observed on dataset II, compared to dataset I, is an expected and reasonable result. It reflects the inherent difficulty of the task and provides a rigorous evaluation of the model's ability to generalize across substantially different viewpoints. Despite the challenge, the proposed model achieves the best performance among all methods, demonstrating its strong generalization capacity under extreme viewpoint variations, as shown in Table 7.

To compare and validate the computational complexity and inference speed of the proposed model against other models, we conducted experiments under consistent conditions. Specifically, we maintained a batch size of 1 and used the same image resolution and keypoint input format as in previous experiments. The computational complexity, measured in FLOPs, was evaluated on an NVIDIA GeForce RTX 2080 Ti GPU. Inference time for all models was measured on a 12th Gen Intel (R) Core (TM) i7-12700F CPU. The results are summarized in Table 9. The experimental results demonstrate that the proposed model maintains a significant advantage over other models in both computational complexity and inference time.

It is worth noting that our experimental results reveal a seemingly counterintuitive phenomenon: in terms of several evaluation metrics, models with more parameters and larger scale underperform compared to some smaller models with fewer parameters. This observation can be explained from two perspectives. First, when considering models with different architectures, model performance is not solely determined by the number of parameters. For structurally different models, factors such as architectural design, optimization strategy, and the alignment between the model structure and the specific dataset often play a more critical role than model size alone (Chen et al., 2022). A smaller model with a compact design and well-tuned optimization may achieve stronger representational power under certain conditions, outperforming larger models with less efficient architectures. Second, even under the same architecture, varying model scales can lead to non-monotonic changes in performance. This phenomenon is explained by the theory of double descent (Nakkiran et al., 2021). As model capacity increases and approaches the point where it can perfectly fit the training data, including noise and outliers, generalization performance may temporarily degrade. At this critical regime, the model may overfit to noisy details, resulting in solutions that are highly sensitive to input perturbations and generalize poorly. However, as the capacity increases further into the overparameterized regime, test error may begin to decrease again. In this stage, the model can exploit its large representational power to learn smoother, more generalizable functions. In our experimental setting, the larger models that perform worse than their smaller counterparts likely fall within this critical regime, where generalization is compromised. In contrast, the smaller models, which have not yet reached this regime, tend to learn solutions with better generalization, resulting in superior test performance.

To enhance the interpretability of the model's results, we visualized and compared the adjacency matrix of the initial 18 keypoint connections defined by OpenPose with the adjacency matrix processed by our proposed LGC module, as shown in Fig. 7. The visualization demonstrates that the original model primarily focuses on local relationships between adjacent keypoints. In contrast, our LGC module significantly expands the receptive field of each keypoint, enabling global interactions across all keypoints. Moreover, the LGC module assigns varying weights to different keypoint pairs, allowing the model to focus more effectively on body parts that are more informative for the recognition task, thereby improving its overall discriminative capability.

To further demonstrate the overall performance of the proposed model

across different tasks, we visualize a comparative analysis of various mainstream models on the MSP I and BPD datasets in terms of classification accuracy, as shown in Fig. 8. The horizontal axis represents accuracy on the MSP I dataset, while the vertical axis indicates accuracy on the BPD dataset. The size of each bubble corresponds to the model's parameter count. As illustrated, the proposed model (marked with a red star) achieves the highest accuracy on both datasets, despite having only 0.1M parameters. This highlights its exceptional balance between high performance and lightweight design. In contrast, although some larger models also demonstrate competitive accuracy, they come with significantly higher parameter counts and computational demands, making them less suitable for resource-constrained or real-time applications.

The test results of the proposed model are shown in Fig. 9, while the test results of the model combined with the recommendation for sitting posture correction are shown in Fig. 10.

Table 6
Comparison with other network models on MSPI.

Method	Pretrained	Params	Accuracy	Precision	Recall	F1-Score
VGG16	Yes	138.36	96.70	97.04	96.39	0.964
SqueezeNet	Yes	1.24	93.75	94.64	93.25	0.936
ResNet50	Yes	25.56	97.44	97.60	97.14	0.972
ResNet101	Yes	44.52	98.18	98.29	97.95	0.979
Xception	Yes	20.82	97.51	97.59	97.23	0.972
ST-GCN	No	3.08	95.50	95.96	95.22	0.953
MobileNetV3	Yes	2.54	95.43	95.54	95.35	0.953
GhostNet	No	3.91	89.58	90.36	89.41	0.894
ViT-B/16	Yes	85.804	95.29	95.73	94.89	0.949
MS-G3D	No	2.99	97.31	97.86	97.07	0.973
Swin-T	Yes	27.52	95.22	95.36	94.95	0.950
CMT-Ti	No	8.43	93.75	94.02	93.35	0.934
CMT-XS	No	15.24	93.145	93.33	92.66	0.927
LMSPNet	No	0.199	92.80	92.76	92.22	0.923
VanillaNet-5	No	17.26	93.95	94.44	93.48	0.936
VanillaNet-6	No	51.05	96.57	96.82	96.29	0.963
VanillaNet-7	No	51.60	96.84	97.03	96.55	0.966
LGCSPNet (ours)	**No**	**0.097**	**99.06**	**99.08**	**99.04**	**0.990**

Method refers to the model architecture. Pretrained indicates whether pretrained weights are used. Params denotes the number of trainable parameters. Accuracy is the overall prediction accuracy, Precision is the proportion of true positives among predicted positives, Recall is the proportion of true positives among actual positives, and F1-Score is the harmonic mean of precision and recall. The definitions of these metrics also apply to Tables 7 and 8.

Table 7
Comparison with other network models on MSP II.

Method	Pretrained	Params	Accuracy	Precision	Recall	F1-Score
VGG16	Yes	138.36	41.52	54.85	42.99	0.377
SqueezeNet	Yes	1.24	33.33	58.02	35.00	0.307
ResNet50	Yes	25.56	36.32	34.09	37.91	0.284
ResNet101	Yes	44.52	42.57	45.99	44.01	0.400
Xception	Yes	20.82	58.47	62.04	59.51	0.525
ST-GCN	No	3.08	55.04	56.70	55.28	0.515
MobileNetV3	Yes	2.54	48.33	48.52	49.14	0.439
GhostNet	No	3.91	37.71	39.35	39.27	0.292
ViT-B/16	Yes	85.80	42.84	58.56	44.26	0.373
MS-G3D	No	2.99	38.64	40.93	40.19	0.294
Swin-T	Yes	27.52	39.93	62.59	41.27	0.333
CMT-Ti	No	8.43	57.63	66.00	58.70	0.611
CMT-XS	No	15.24	60.00	67.64	61.00	0.636
LMSPNet	No	0.199	54.69	60.74	55.48	0.570
VanillaNet-5	No	17.26	65.13	68.14	66.01	**0.669**
VanillaNet-6	No	51.05	64.23	68.64	65.12	0.666
VanillaNet-7	No	51.60	63.61	**69.03**	64.52	0.664
LGCSPNet (ours)	**No**	**0.097**	**65.62**	59.25	**66.49**	0.605

Table 8
Comparison with other network models on BPD.

Method	Pretrained	Params	Accuracy	Precision	Recall	F1-Score
VGG16	Yes	138.36	97.58	98.21	97.94	0.978
SqueezeNet	Yes	1.24	95.24	95.86	95.21	0.951
ResNet50	Yes	25.56	98.98	99.16	99.13	0.991
ResNet101	Yes	44.52	98.65	98.74	98.43	0.986
Xception	Yes	20.82	95.63	96.76	96.06	0.957
ST-GCN	No	3.08	97.74	98.20	97.87	0.978
MobileNetV3	Yes	2.54	91.57	93.38	91.59	0.916
GhostNet	No	3.91	89.85	92.27	90.49	0.894
ViT-B/16	Yes	85.804	98.82	98.88	98.73	0.987
MS-G3D	No	2.99	99.38	99.43	99.47	0.994
Swin-T	Yes	27.52	97.26	97.50	97.35	0.972
CMT-Ti	No	8.43	95.47	96.84	96.16	0.957
CMT-XS	No	15.24	91.65	93.92	92.88	0.910
LMSPNet	No	0.199	89.14	91.75	89.49	0.889
VanillaNet-5	No	17.26	91.70	95.39	92.90	0.915
VanillaNet-6	No	51.05	94.93	96.63	95.70	0.952
VanillaNet-7	No	51.60	94.94	96.66	95.74	0.952
LGCSPNet (ours)	**No**	**0.097**	**99.92**	**99.91**	**99.90**	**0.999**

4.4. Ablation Study

In this section, we verify the effectiveness of specific components of the proposed model by removing them individually. Specifically, for contrastive learning and parameter-free attention, we directly eliminate them during training; for NSG, we replace it with a 1×1 convolution. The experimental results are shown in Table 10.

The experimental results demonstrate that incorporating contrastive learning (CL), parameter-free attention, and the non-local shift graph operation (NSG) significantly enhances prediction accuracy across multiple datasets. Specifically, these modules improve accuracy on MSP I by 1.88%, 2.39%, and 0.81%, respectively; on MSP II by 11.25%, 12.15%, and 7.22%, respectively; and on the BPD by 4.99%, 6.63%, and 2.88%, respectively. Notably, all three modules exhibit the most substantial improvements on the MSP II dataset. This pronounced effect may be attributed to the lower baseline model performance, which results from the dataset's more complex perspective variations

Table 9
FLOPs and inference time.

Method	GFLOPs	Inference time (ms)
VGG16	15.47	244.32
SqueezeNet	0.35	13.18
ResNet50	4.11	72.05
ResNet101	7.84	131.41
Xception	4.57	93.86
ST-GCN	0.06	1.89
MobileNetV3	0.06	2.68
GhostNet	0.14	9.07
ViT-B/16	17.60	277.27
MS-G3D	0.11	5.06
Swin-T	4.52	118.82
CMT-Ti	0.64	41.30
CMT-XS	1.54	66.05
LMSPNet	0.14	11.45
VanillaNet-5	5.20	155.53
VanillaNet-6	6.01	187.23
VanillaNet-7	6.89	214.80
LGCSPNet (ours)	0.001	0.37

GFLOPs indicates the number of floating-point operations required for a single forward pass, serving as a measure of computational complexity. Inference time is the average time taken to process one sample on the target hardware.

Table 10
Ablation studies.

NSG	Parameter-Free Attention	CL	MSP I (%)	MSP II (%)	BPD (%)
✓	✓		97.18	54.37	94.93
✓		✓	96.77	53.47	93.29
	✓	✓	98.25	58.40	97.04
✓	✓	✓	99.06	65.62	99.92

NSG, Parameter-Free Attention, and CL represent the three modules composing the model. MSP I, MSP II, and BPD are the model's accuracy metrics (%) on different benchmark datasets.

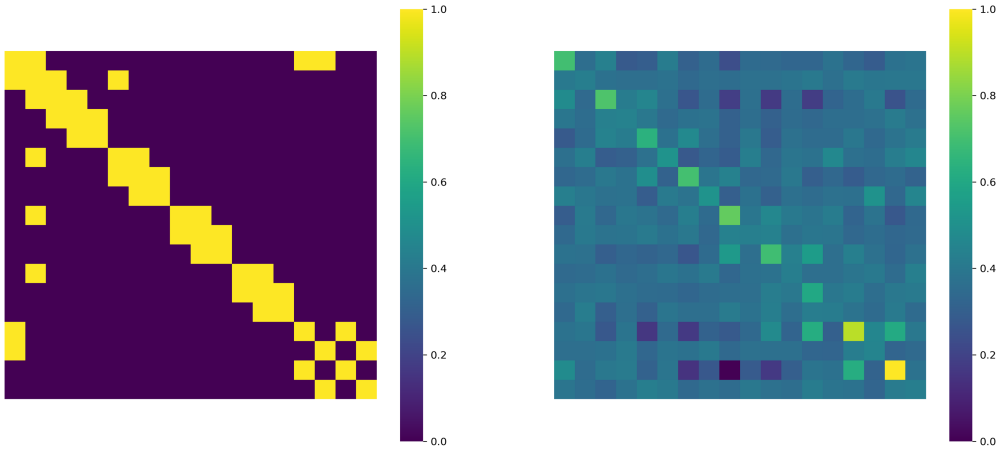

Fig. 7. The visualization of the adjacency matrix: The left image shows the initial adjacency matrix, while the right image shows the adjacency matrix after processing by the LGC module.

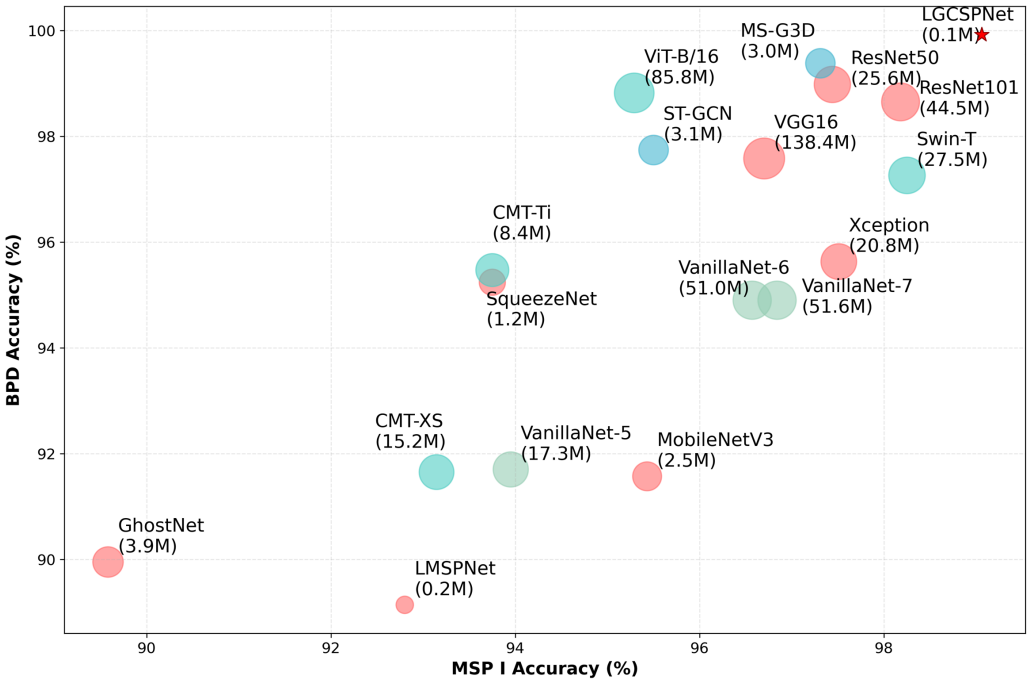

Fig. 8. Test comparison of LGCSPNet and mainstream models.

Fig. 9. Test results (display the identified category and confidence level in the top-left corner of the image).

and higher task difficulty. Consequently, the need for enhanced feature focus and global modeling is more critical, allowing these modules to yield greater performance gains. From a module-specific perspective, the parameter-free attention mechanism consistently delivers the most significant improvements across all three datasets. This suggests that it effectively enables the model to concentrate on key features, thereby enhancing the model's ability to discern complex postures. CL and NSG also contribute meaningful improvements,

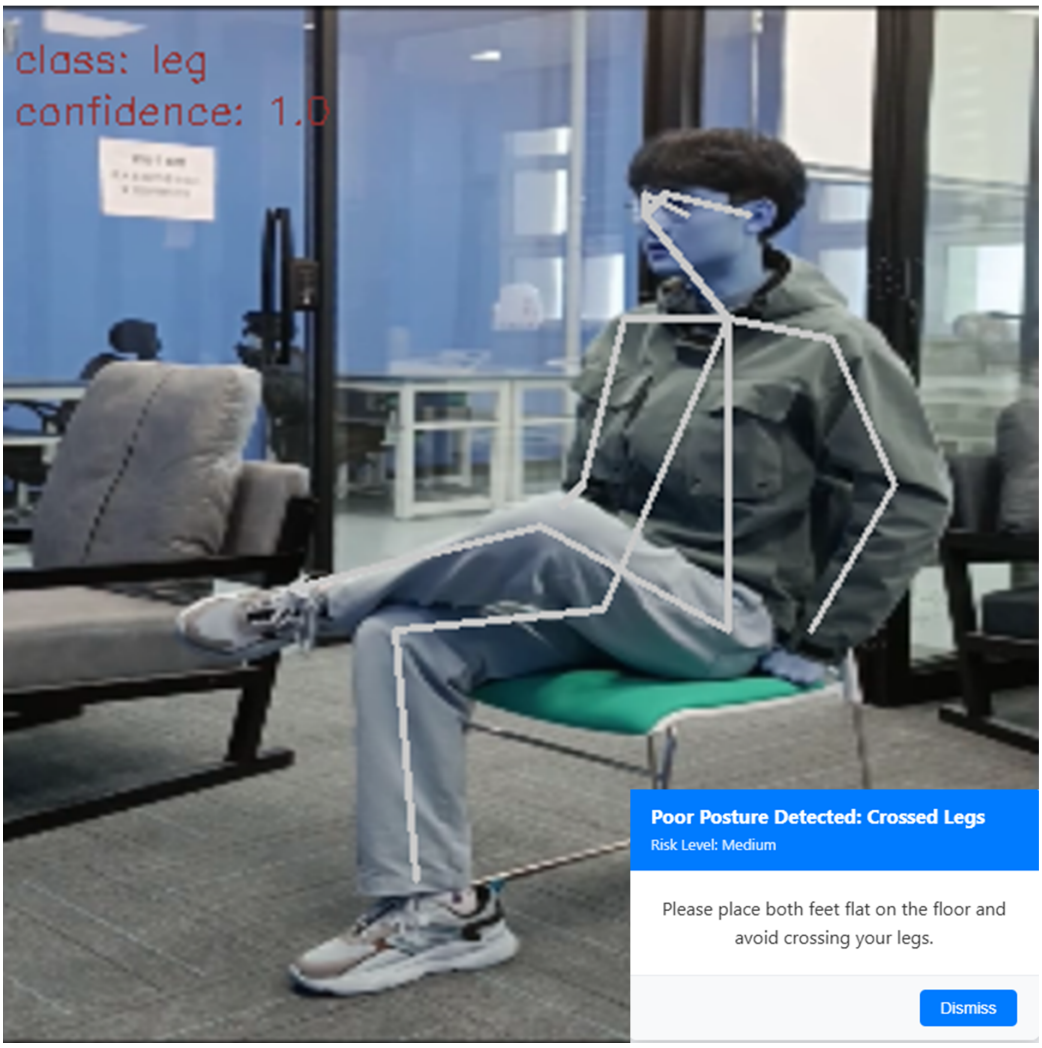

Fig. 10. Posture adjustment recommendations for leg-crossing.

albeit to varying degrees. This suggests that CL strengthens the consistency of features among intra-class views and reduces the similarity of features between inter-class views, while NSG enhances the model's understanding of the overall posture structure by capturing global dependencies. When integrated, these three modules achieve the highest overall performance, demonstrating a strong synergy among them. This combined effect highlights the indispensability of each module and substantiates the rationality and necessity of their design in optimizing model performance.

Table 11
Effect of the number of LGC modules.

Number of LGC	Params	MSP I (%)	BPD (%)	MSP II (%)
1	4623	84.27	89.23	61.60
2	41023	95.36	91.4	63.26
3	**97119**	**99.06**	**99.92**	**65.62**
4	101783	97.31	96.33	62.64
5	102963	97.64	93.83	60.35
6	103265	95.69	90.63	59.93

Number of LGC indicates the number of distinct LGC modules included in the model. Params represents the number of parameters in the model, reflecting its complexity. MSP I, BPD, and MSP II represent the model's testing accuracy on different datasets, respectively.

4.5. Effect of the Number of LGC Modules on Model Performance

We study the number of LGC modules in the model to explore its impact on model performance. The experimental results are shown in Table 11. For the LGC module, experimental results indicate that three layers of LGC blocks can achieve the best performance. The reason may be that with fewer than three layers, the model underperforms due to insufficient depth. A single layer connects distant nodes but lacks the capacity to abstract these features, while two layers begin integrating global information yet fail to capture complex, multi-level patterns in the data. In contrast, exceeding three layers degrades performance as the highly efficient non-local mechanism leads to over-smoothing coupled with increased computational cost and overfitting risks. Three layers achieve an optimal balance: the first layer establishes long-range dependencies, the second refines them into a cohesive representation, and the third optimizes these features for downstream tasks.

5. Discussions

In this work, several assumptions underpin the design and evaluation of LGCSPNet. First, the model presumes that posture categories can be effectively distinguished based on coordinate-space representations, a capability further reinforced through contrastive learning. While this approach exhibits strong generalization across standard conditions, its performance in more complex deployment scenarios could benefit from the integration

of additional contextual cues, such as ergonomic principles or task-specific constraints, to refine decision boundaries in cases involving subtle postural variations. Furthermore, current evaluations are primarily conducted on public datasets and a custom dataset collected from adult participants in typical office settings. To ensure broader applicability, future research may extend the evaluation to more diverse user groups, including children, older adults, or individuals with mobility impairments, thereby assessing the model's robustness across different demographic and physiological conditions.

Despite its strengths, the proposed model faces certain limitations. Like most skeleton-based methods, its performance heavily depends on the quality of input coordinates obtained from upstream pose estimation algorithms. Additionally, its lightweight design, while improving computational efficiency, may limit the model's ability to capture fine-grained posture changes when compared to more complex architectures.

Nonetheless, the model offers significant advantages by directly using body keypoint coordinates as input. This design enhances robustness to lighting variations and background noise, while also reinforcing user privacy. Specifically, the model only requires skeletal keypoints for posture recognition, eliminating the need to process raw images or skeleton visuals, thereby reducing the risk of privacy breaches. Moreover, in contrast to traditional, more complex models, the proposed method significantly lowers computational costs, achieving a strong balance between recognition accuracy and computational efficiency, making it suitable for deployment on resource-constrained devices. Experimental results indicate that the model maintains reliable recognition performance while offering real-time responsiveness.

LGCSPNet can be applied to a variety of real-world scenarios. In health management, it enables real-time posture feedback to help prevent musculoskeletal issues associated with prolonged poor sitting habits. In driver monitoring systems, it can detect postures linked to fatigue or inattention, thereby assisting in assessing driver status and supporting road safety. In smart home and healthcare settings, the model can contribute to elderly care and rehabilitation by recognizing diverse postures. In human-computer interaction, real-time posture recognition can serve as a novel, natural input method, enhancing user experience in controlling smart devices and interacting with virtual environments. In conclusion, this study presents a promising lightweight posture recognition method that establishes a solid foundation for practical applications and outlines a clear direction for future innovation.

6. Conclusions

In this research, we propose a novel lightweight graph convolutional sitting posture recognition network, LGCSPNet, which is used to identify bad sitting postures and reduce the harm of bad sitting postures to the human body. By adopting the lightweight NSG and SimAM, we construct the LGC module, enabling global modeling of human skeletons and attention to key regions while maintaining low resource requirements. To better explore interclass relationships, we introduce contrastive learning. Testing results on the multiview sitting posture dataset and body pose dataset demonstrate that the proposed network achieves the highest recognition accuracy with the smallest number of parameters. Ablation experiments further verify the effectiveness and synergy of each module in the network. Experimental results on the multiview sitting posture dataset II show that the model also performs outstandingly in cross-view recognition compared to other networks, validating the cross-view generalization capability of the proposed network. Furthermore, we refer to ergonomic knowledge and perform risk classification on the identified sitting postures in order to better guide the improvement of sitting postures and reduce the harm of bad sitting postures.

The proposed system has demonstrated initial success in real-time detection of poor sitting postures and providing corrective suggestions, showing strong application potential. Future work will focus on three key areas: data, model, and functionality. On the data side, we plan to construct a more diverse and representative dataset covering a wider range of ages, body types, and sitting behaviors to enhance generalization. On the model side, we aim to explore more efficient and robust pose recognition and classification algorithms to improve detection accuracy and responsiveness. On the functionality side, we will refine the personalized recommendation mechanism by incorporating user feedback for adaptive strategy optimization, and introduce health reports, goal-setting features, and multimodal interactive feedback to improve user engagement and intervention effectiveness. Through these continuous iterations, the system is expected to evolve into an intelligent and practical solution for healthy sitting posture management.

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